

CHAPTER 2

METHODOLOGY - MARKED REVISION

Color key: Revised wording New wording or revised figure Removed wording or obsolete figure

This is a thesis draft with visible revisions. LEAPRS system behavior follows the system concept skill. Study dates, participant recruitment, and team practices require confirmation from the researchers.

Introduction

This chapter presents the research methods, procedures, and tools employed in the development and evaluation of LEAPRS: An AI-Assisted Web-Based Training Requisition and Monitoring System for Pamantasan ng Lungsod ng Pasig. It outlines the research design, the development approach, the intended respondents, and the data collection instruments. These methods are intended to connect the administrative problems described in Chapter 1 with a systematic account of how the system was built and how it will be assessed.

2.2 Research Design

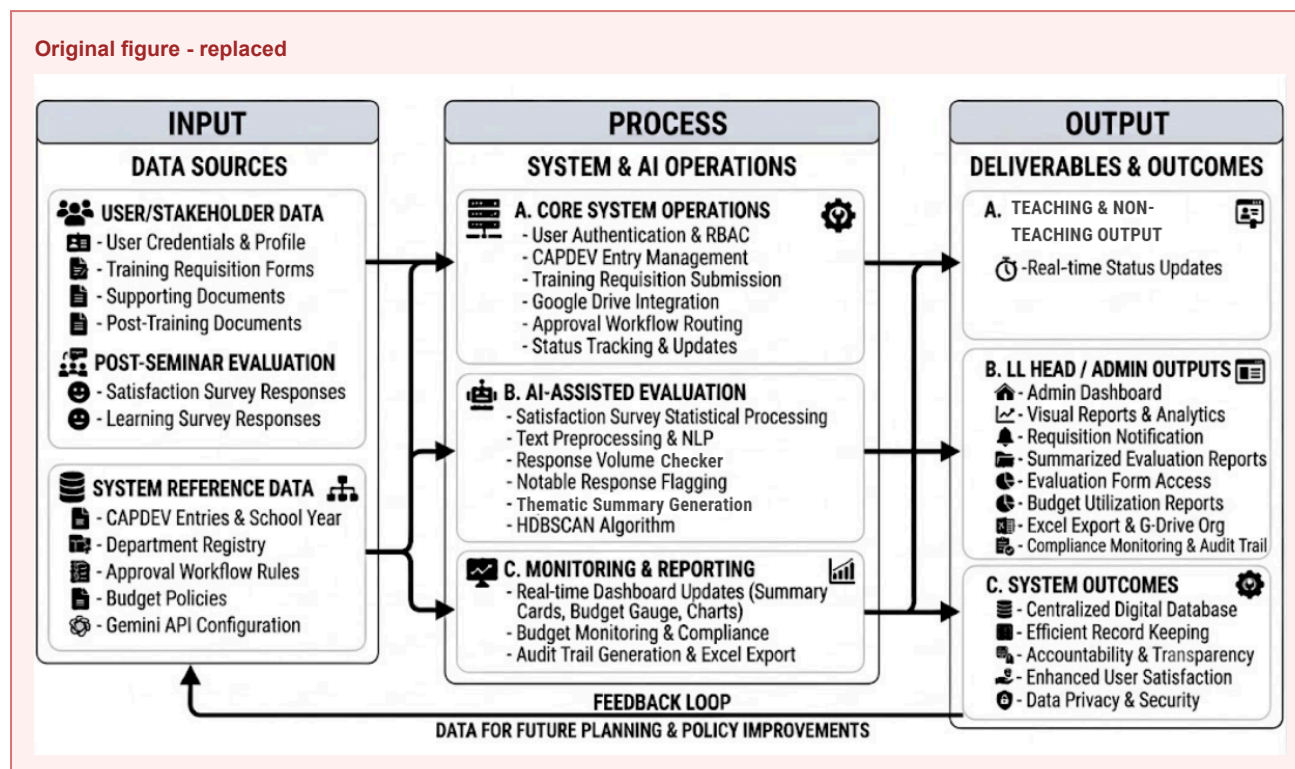
~~The original chapter described an implemented HDBSCAN-based AI analysis as part of the developmental study and treated the evaluation as completed.~~

This study uses a developmental research design to examine the design, development, and evaluation of LEAPRS. Richey and Klein (2005) describe developmental research as a systematic study of products and development processes. Dela Rosa (2023) provides a related example of developing a web application and evaluating it with software-quality criteria. In this study, the system is developed around the training requisition needs identified at PLP, and its completed functions are then evaluated by intended users.

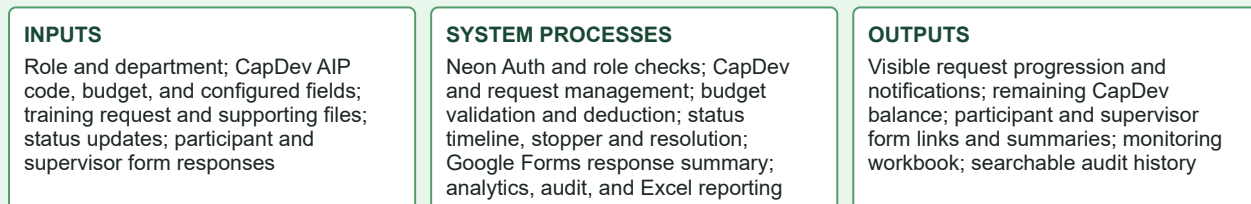
The user evaluation is quantitative. Respondents will complete a structured questionnaire using a five-point Likert scale, and responses will be summarized using weighted means for each criterion. The questionnaire follows the eight product-quality characteristics of ISO/IEC 25010:2011 so it remains aligned with the research questions in Chapter 1. That edition has been replaced by ISO/IEC 25010:2023, which contains nine characteristics; the researchers should state clearly which edition their instrument follows. Questionnaire results express respondents' perceptions and do not, by themselves, establish measured security, reliability, or maintainability.

2.2.1 Architectural Framework

Figure 1. Architectural Framework of LEAPRS



Revised Figure 1. Architectural Framework of LEAPRS



External services: Neon Postgres/Auth, Google Drive/Forms, Gemini for help and request drafting, Groq for written-feedback summaries.

Figure 1 follows the movement of information from request submission to monitoring and evaluation. Users supply CapDev and request information, attachments, status updates, and evaluation responses. LEAPRS checks role and department access, stores application records in Neon Postgres, and keeps attachment files in Google Drive. It records the status timeline and budget deductions, then presents the resulting information through request views, notifications, analytics, audit history, and Excel monitoring reports.

After a request is explicitly completed, LEAPRS creates a participant evaluation form and a supervisor or post-activity evaluation form. Rating charts are calculated from the latest responses when See Summary is opened, while Groq may summarize written comments if configured. Gemini serves separate help-chat and activity-design draft functions. The framework does not include automatic office-to-office approval routing, HDBSCAN clustering, or an automatically updated compliance score.

Evaluation Processing Procedure

~~Algorithm Discussion: LEAPRS uses HDBSCAN, all MiniLM L6-v2 sentence embeddings, Euclidean pairwise distances, core distances, mutual reachability, a minimum spanning tree, and membership scores before Gemini generates each theme summary.~~

~~Tables 1-3: six sample responses, their pairwise Euclidean distance matrix, and final HDBSCAN cluster assignments and membership strengths. These tables illustrate a pipeline that LEAPRS does not implement.~~

The implemented post-activity evaluation flow begins when an authorized user concludes a request as Completed. LEAPRS creates two Google Forms unique to that request: one for participant feedback and one for supervisor or post-activity evaluation. The request stores the form IDs and responder links so the forms can be opened and shared from the completion view. Responses are submitted through Google Forms by people with the links.

When an authorized user selects See Summary, LEAPRS fetches the current form responses. It calculates rating distributions and averages from the scale-based answers. If Groq is configured, the application sends written feedback for a concise summary of strengths, areas for improvement, and recommended actions. If AI summarization fails, the rating charts remain available. This on-demand flow uses neither text embeddings nor a clustering algorithm.

The generated text should be interpreted as an aid to review. The researcher or Lifelong Learning Head should inspect the underlying comments before drawing conclusions about training effectiveness, especially when the number of responses is small or comments differ substantially.

2.2.3 Software Engineering Paradigm

Figure 2. Scrum Cycle

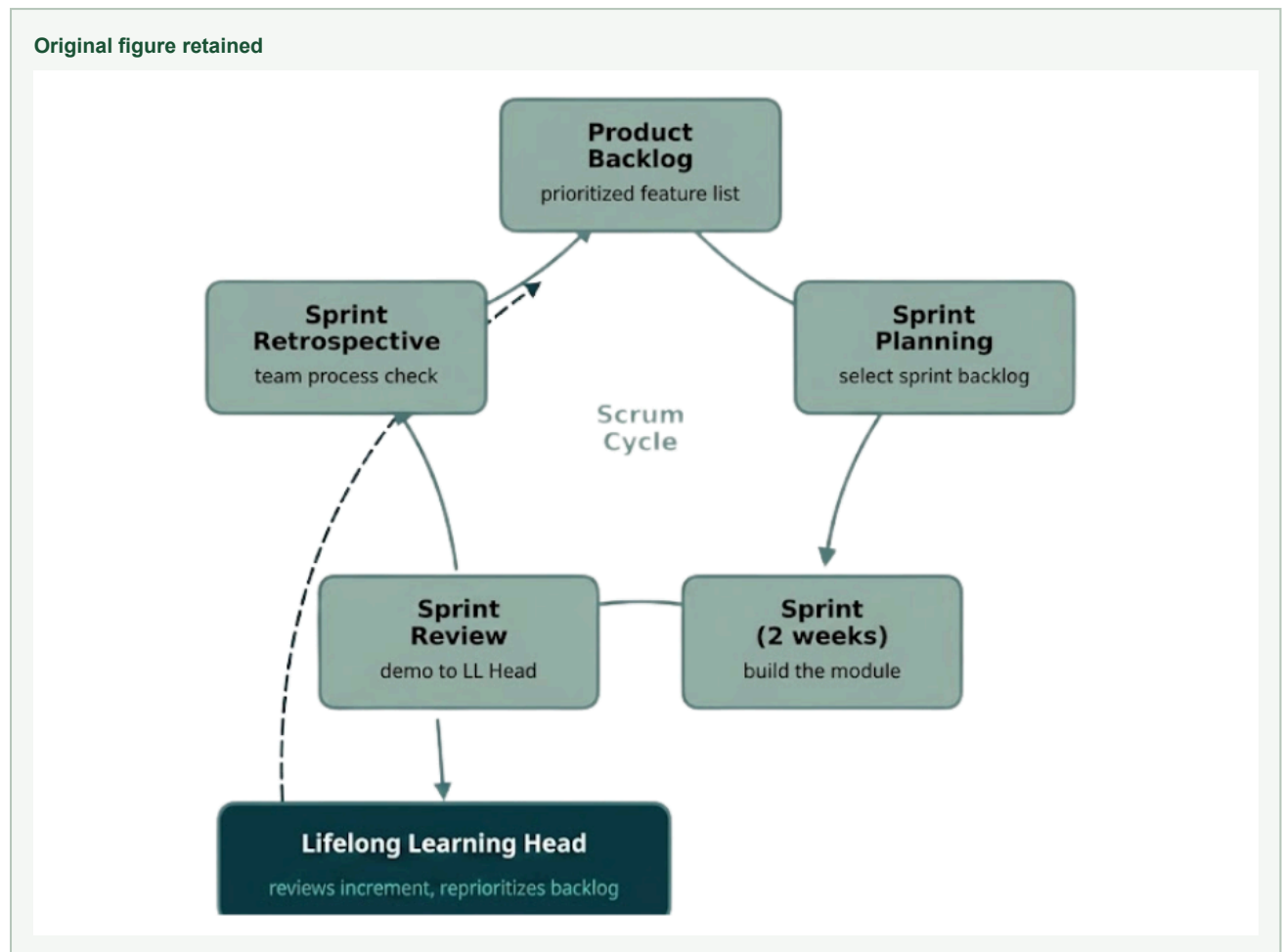


Figure 2 remains a suitable conceptual Scrum-cycle illustration. The exact two-week cadence, the named Product Owner and Scrum Master, and the number of completed sprints must be checked against team records before this diagram is presented as the process actually followed.

Rationale for Adopting Agile

LEAPRS was developed through iterative work because its requirements span several connected activities: CapDev configuration, employee requests, document uploads, budget controls, status timelines, evaluations, and reporting. Feedback from the Lifelong Learning Head and other intended users can be incorporated into later increments after a working feature has been demonstrated. This approach is consistent with a developmental study in which the design is refined as its fit with the administrative setting becomes clearer.

The application can be divided into functional increments without treating each one as an isolated final product. For example, CapDev and request records establish the data on which status monitoring and reporting depend, while the post-activity forms are triggered only after a request is completed. Iterative integration and testing help reveal mismatches among these flows before the final evaluation.

Scrum Framework and Team Roles

Scrum provides a structure for planning work, reviewing a usable increment, and adapting the backlog. The 2020 Scrum Guide identifies three accountabilities: Product Owner, Scrum Master, and Developers. For this capstone, the researchers should document which person actually held each accountability, how the Lifelong Learning Head participated as a stakeholder, and the dates and outcomes of the sprints. An adviser or stakeholder should be named Product Owner only if that person actually managed the Product Backlog and product priorities.

Sprint Breakdown

~~Table 4 assigns a sprint to automatic approval workflow routing, post-training compliance automation, response embedding, HDBSCAN clustering, and Gemini-generated theme summaries.~~

Table 4. Revised Sprint Focus (proposed structure; verify against project records)

Sprint	Focus
0	Requirements, stakeholder review, backlog, architecture and database design
1	Neon Auth, roles and departments, CapDev records and form configuration
2	Training request creation, configured fields, supporting-file upload
3	Request status timeline, stopper/resume, completion/denial, notifications
4	Budget checks, one-time deduction, CapDev balance history, audit history
5	Participant and supervisor Google Forms, on-demand response summaries
6	Analytics, monitoring workbook, Gemini help and activity-design draft
7	Integration, access checks, user testing, refinement, evaluation preparation

Table 4 aligns the development increments with the implemented modules of LEAPRS. It is a proposed presentation of the work, not proof that these tasks occurred in this order. The final thesis should replace the proposed sequence with dated sprint records, backlog items, review notes, and completed deliverables.

Scrum Ceremonies

Sprint Planning selects the work for an iteration. During development, the team checks progress and raises blockers. A Sprint Review provides a point for the Lifelong Learning Head to examine the working increment and suggest changes, while a Retrospective helps the researchers improve their process. The thesis should describe only the ceremonies that were actually held and cite meeting notes or sprint artifacts where available.

~~The iterative structure includes an empirical comparison of HDBSCAN, DBSCAN, and OPTICS in a dedicated sprint; the benchmark results determine the final algorithm.~~

The iterative structure also accommodates changes in external-service integration, such as Google Forms, Google Drive, and AI provider configuration. These concerns can be tested and refined without adding an unimplemented clustering benchmark to the research method.

Figure 3. Level 0 Data Flow Diagram of LEAPRS

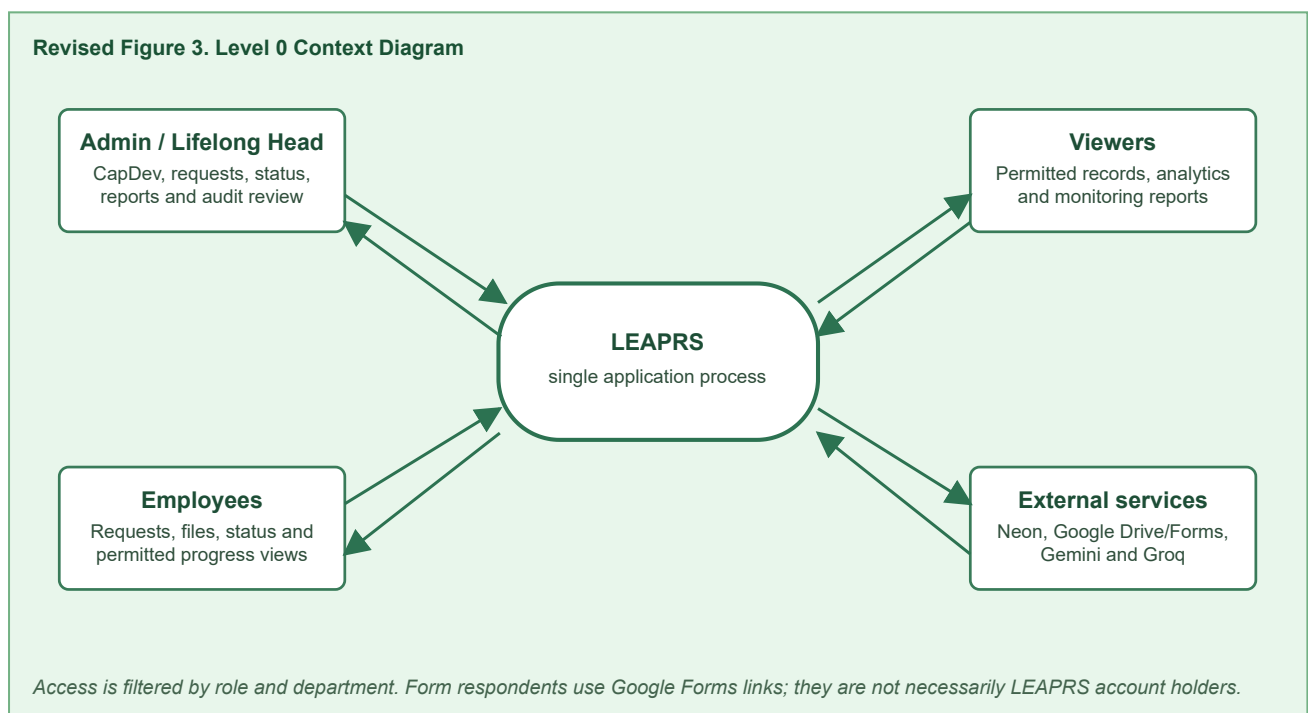
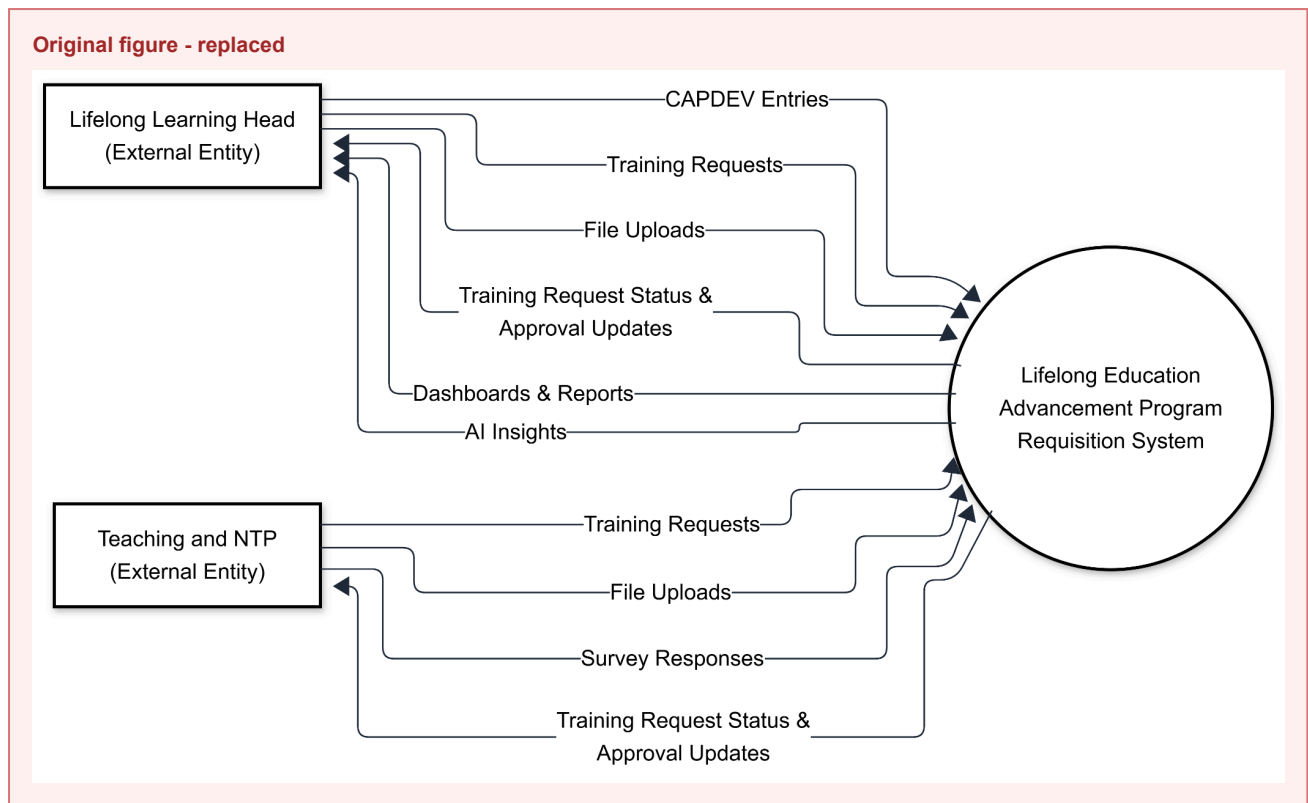
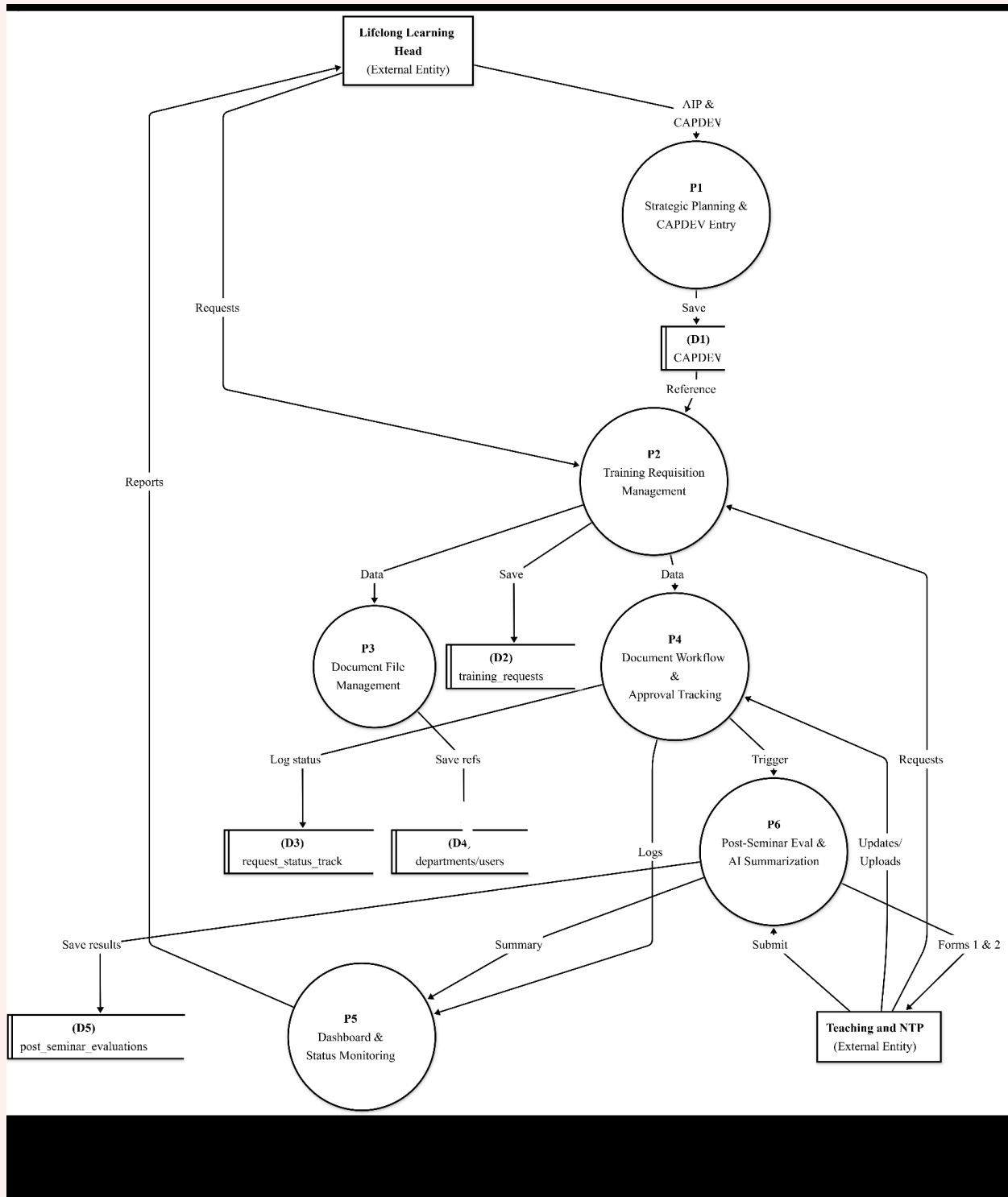


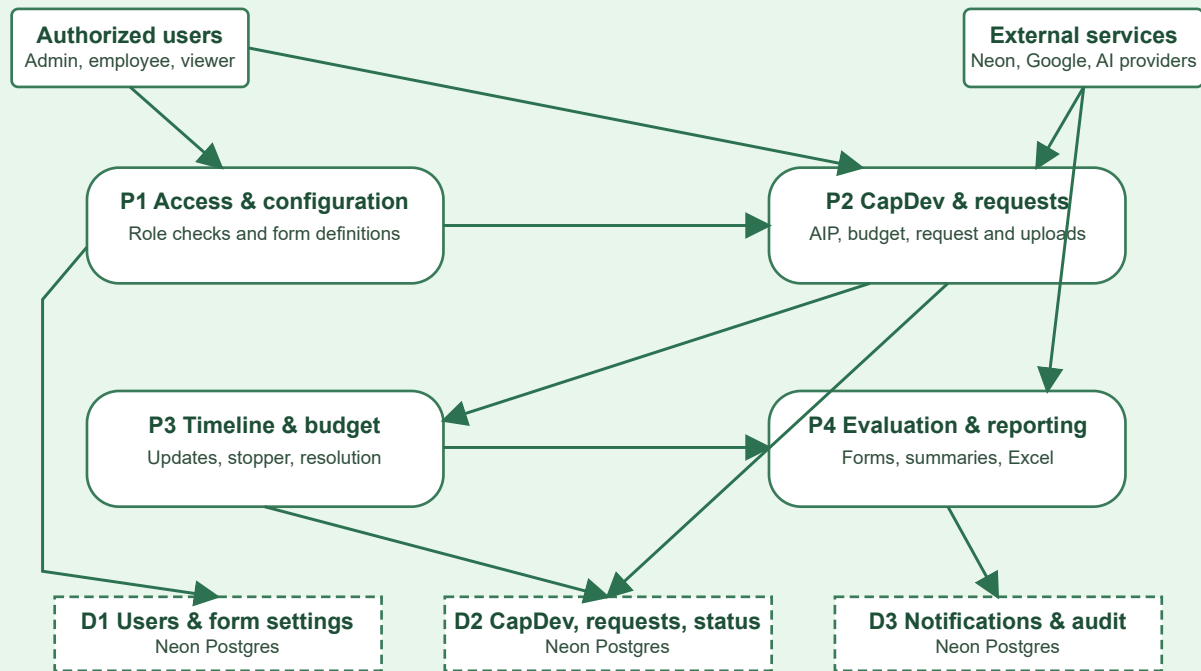
Figure 3 presents LEAPRS as a single system interacting with authorized internal roles and external services. Admins configure and monitor the system, employees submit and follow requests within their permissions, and Viewers inspect permitted records and reports. Google Drive stores attachments, Google Forms collects post-activity responses, Neon provides authentication and application data storage, and AI services support separate help, drafting, and feedback-summary functions. The figure no longer implies that LEAPRS physically routes documents through approving offices.

Figure 4. Level 1 Data Flow Diagram of LEAPRS

Original figure - replaced



Revised Figure 4. Level 1 Data Flow Diagram

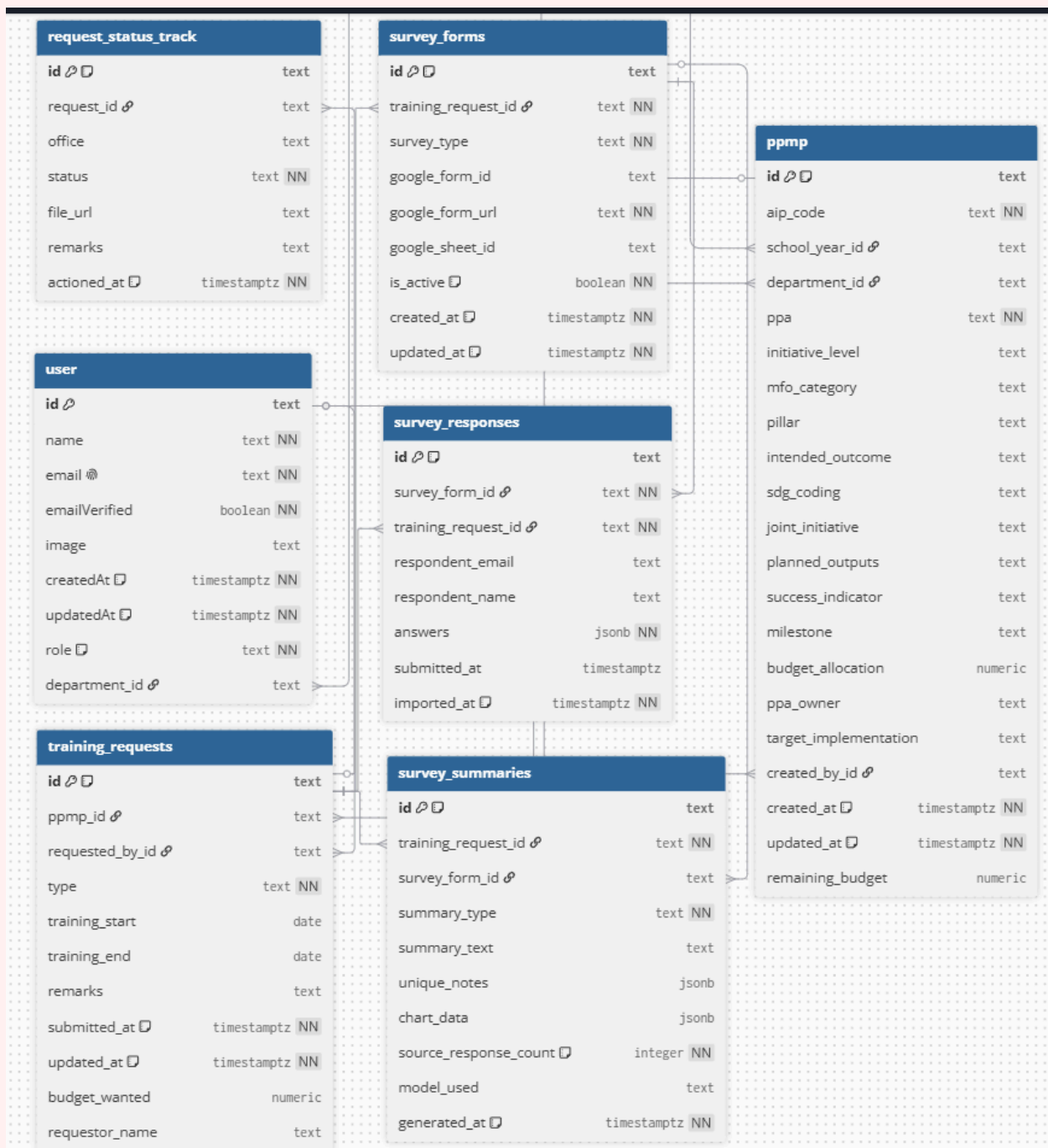


Cross-cutting records: notifications and audit history. Durable files: Google Drive. Application records: Neon Postgres.

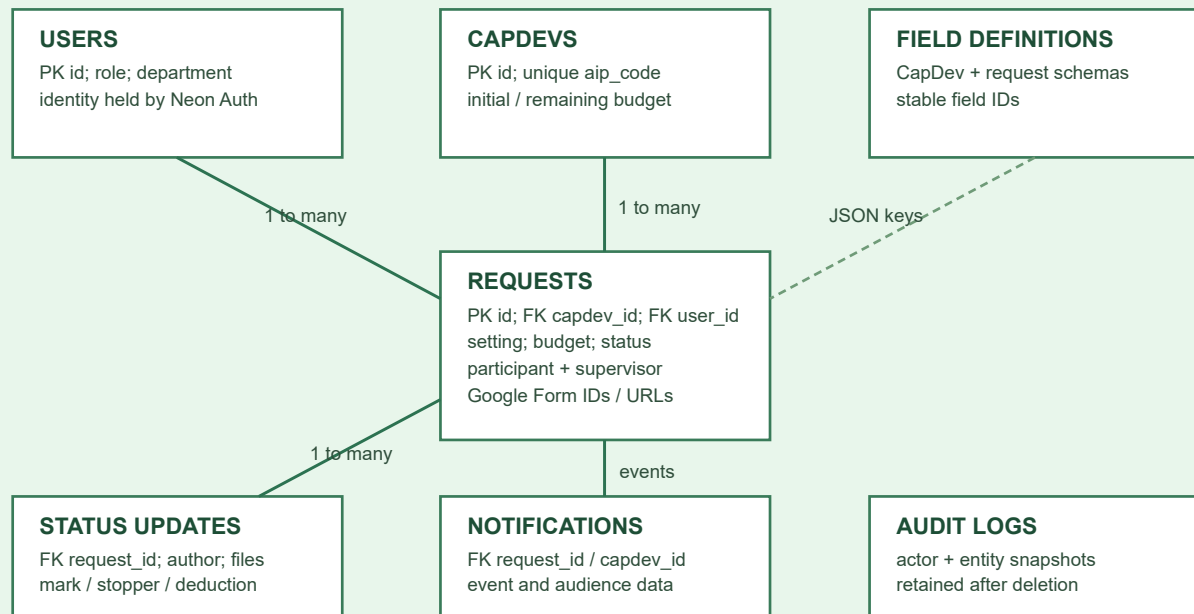
Figure 4 decomposes LEAPRS into access and configuration, CapDev and request management, timeline and budget handling, and evaluation and reporting. A request references a CapDev project; its timeline records progress, stoppers, and an explicit final resolution. Completion creates the two evaluation forms. The application retrieves responses when a summary is requested, while notifications and audit records document relevant activity. This flow replaces the obsolete diagram's PPMP, office-routing, Gemini survey-analysis, and survey-response-table paths.

Figure 5. Core Data Model of LEAPRS

Original figure - replaced



Revised Figure 5. Core Entity Relationships



No PPMP, school_years, departments registry, survey_responses, or survey_summaries tables appear in the current application schema.

Figure 5 presents the main persisted entities used in the current implementation. A CapDev project is identified to users by its unique AIP Code and can have many associated requests. Each request belongs to a user and can have many status updates. Configured form values are stored in JSON fields against stable field-definition IDs. The request stores identifiers and links for its participant and supervisor Google Forms. LEAPRS retrieves form responses on demand; it does not store them in the survey response and summary tables shown in the earlier diagram.

User profiles hold application roles and departments, while Neon Auth manages identity and sessions. Notifications are tied to relevant records and audience rules. Audit entries retain actor and record snapshots after the related business record is removed. Google Drive contains attachment bytes; the database stores references and metadata. The complete physical schema has additional operational tables, so Figure 5 is a core-entity view rather than a claim that LEAPRS has exactly nine tables.

2.3 Population and Sampling

The population of the study consists of PLP personnel involved in training requisition submission and oversight. The proposed pilot includes teaching personnel from the College of Computer Studies, non-teaching personnel from administrative offices, and the Lifelong Learning Head. These groups represent the employee and administrative perspectives needed to evaluate the workflow described in Chapter 1.

Purposive sampling is appropriate because respondents are selected for their involvement in training requisitions and their likely use of LEAPRS. The draft proposes eleven respondents: five teaching personnel, five non-teaching personnel, and the Lifelong Learning Head. Before final submission, the researchers should confirm the actual recruited count, eligibility criteria, and whether all eleven completed the evaluation.

Teaching Personnel of the College of Computer Studies. These participants evaluate the request form, document submission, recorded progress, and notifications from the perspective of faculty requestors.

Non-Teaching Personnel. These participants evaluate the same workflow from administrative and support roles, whose training purposes may differ from those of faculty.

Lifelong Learning Head. The Head evaluates the usefulness of CapDev and budget records, request histories, post-activity forms, reporting, and feedback summaries for oversight. The assessment should concern the functions actually available in LEAPRS, rather than automatic office approval routing or machine-learning theme discovery.

2.4 Data Collection Method

The study will use a structured questionnaire organized around the eight product-quality characteristics of ISO/IEC 25010:2011: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The chapter and questionnaire should use these formal characteristic names consistently. The chosen 2011 edition should be stated explicitly because ISO/IEC 25010:2023 has a revised nine-characteristic model.

Respondents will interact with LEAPRS using representative tasks and then answer a five-point Likert questionnaire, from Strongly Disagree (1) to Strongly Agree (5). Responses may be summarized with weighted means for each item and criterion. The actual task list, questionnaire items, interpretation ranges, and participant counts should be included in the thesis appendices so the evaluation can be checked and repeated.

User ratings can indicate perceived quality, but some characteristics require other evidence. For example, security claims should be supported by role-access checks, reliability by observed failures or recovery tests, and maintainability by developer review of the code and change process. If only the respondent survey is conducted, findings should be reported as perceived quality rather than objective verification of those characteristics.

Data collection has three phases: informed consent and preparation, hands-on interaction with the system followed by the questionnaire, and consolidation of responses for analysis. The researchers should record the version and environment of LEAPRS used in the evaluation so the findings can be related to the tested system.

Ethical Considerations

Participation in the evaluation will be voluntary. Respondents will be informed of the study purpose, the activities requested of them, the data being collected, and their right to withdraw without penalty. Consent will be obtained before evaluation begins.

The researchers will avoid publishing names, account details, or training documents in the thesis. Questionnaire data should be stored with restricted access and reported in aggregate or with anonymous participant codes. Any retention or disposal period should follow PLP's applicable institutional requirements and the consent materials provided to participants.

Because LEAPRS uses external services, the evaluation protocol should explain when files or responses pass through Google Drive and Google Forms and when AI requests may be sent to Gemini or Groq.

Demonstrations can use test records where possible. The researchers should avoid sending confidential participant information to AI services unless that use is covered by the institution's approval and participant consent.

The instruments will focus on the performance and usability of LEAPRS without asking intrusive or irrelevant questions. The research team will acknowledge all sources and report the results honestly, including system limitations and any unsuccessful tasks encountered during evaluation.

Source and figure verification notes

All five embedded figures were extracted and inspected. Figures 1, 3, 4, and 5 are replaced above because they depict outdated workflows or database tables. Figure 2 is readable and retained as a conceptual Scrum cycle; its project-specific timing and roles still need team evidence.

- Richey and Klein (2005), Developmental research methods: [DOI 10.1007/BF02961473](https://doi.org/10.1007/BF02961473).
- Dela Rosa (2023), Development of a Web Application for Learning Basic Mandarin Chinese: [DOI 10.3991/ijet.v18i04.37121](https://doi.org/10.3991/ijet.v18i04.37121).
- ISO/IEC 25010:2011, eight-characteristic product-quality model: [official ISO record](#).
- ISO/IEC 25010:2023, revised nine-characteristic model: [official ISO record](#).
- Schwaber and Sutherland (2020), The Scrum Guide: [official guide](#).

Researcher confirmations still needed: actual Scrum cadence and role assignments; sprint dates and artifacts; recruited sample and completed questionnaires; PLP process descriptions; evaluation instrument and task list; data handling approvals.